
Chapter III

Simulation of diffusion processes

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Framework

We consider $D \subset \mathbb{R}$ and the general SDE :

$$dX_t = b(X_t)dt + \sigma(X_t)dW_t, \quad X_0 = x \in D$$

We also define the convenient $\tau_X := \inf\{t : X_t \notin D\}$, $\eta \in \mathbb{R}$ and

$$\begin{cases} N \sim \mathcal{P}(\eta) \\ U_i \sim \mathcal{U}([0, 1]) \forall i \in \llbracket 1, N \rrbracket \end{cases} \text{ independant random variables}$$

1. An new formula for $\mathbb{E}^{\mathbb{P}} \left[f(X_T) \mathbb{1}_{\{\min_T X > L\}} \right]$

Let's start by noticing that $\mathbb{E}^{\mathbb{P}} \left[f(X_T) \mathbb{1}_{\{\min_T X > L\}} \right] = \mathbb{E}^{\mathbb{P}} \left[f(X_T) \mathbb{1}_{\tau_X > T} \right]$ by definition of τ_X . We now want to derive a new expression for this fair-price that would use N and the U_i .

In Chapter I, we have shown that

$$\mathbb{E}^{\mathbb{P}} \left[f(X_T) \mathbb{1}_{\tau_X > T} \right] = \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y_T)) \mathbb{1}_{\tau_Y > T} \cdot e^{h(Y_T) - h(y)} e^{\int_0^T c(Y_s) ds} \right]$$

where $\tau_Y := \inf\{t : Y_t \notin g(D)\}$, $y := Y_0$, $\mathbb{Q}$ is a probability space defined in Chapter I and c is a function defined in Chapter I too

First, we would like to prove that

$$\exp \left(\int_0^T c(Y_s) ds \right) = \mathbb{E}^{\mathbb{Q}} \left[e^{\eta} \prod_{i=1}^N \left\{ \frac{T}{\eta} c(Y_{TU_i}) \right\} \middle| Y \right]$$

Indeed :

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}} \left[e^{\eta \prod_{i=1}^N \left\{ \frac{T}{\eta} c(Y_{Tu_i}) \right\}} \middle| Y \right] &= \sum_{n=0}^{+\infty} \mathbb{Q}(N = n) \cdot \mathbb{E}^{\mathbb{Q}} \left[e^{\eta \prod_{i=1}^n \left\{ \frac{T}{\eta} c(Y_{Tu_i}) \right\}} \middle| Y, N = n \right] \\
&= \sum_{n=0}^{+\infty} \mathbb{Q}(N = n) \cdot e^{\eta \prod_{i=1}^n \left\{ \frac{T}{\eta} \mathbb{E}^{\mathbb{Q}} \left[c(Y_{Tu_i}) \middle| Y, N = n \right] \right\}} \\
&= \sum_{n=0}^{+\infty} \mathbb{Q}(N = n) \cdot e^{\eta \prod_{i=1}^n \left\{ \frac{T}{\eta} \int_0^1 c(Y_{Tu_i}) du_i \right\}}
\end{aligned}$$

where we can apply the change of variable : $s_i = Tu_i$

$$\begin{aligned}
&= \sum_{n=0}^{+\infty} \mathbb{Q}(N = n) \cdot e^{\eta \prod_{i=1}^n \left\{ \frac{T}{\eta} \int_0^T \frac{1}{T} c(Y_{s_i}) ds_i \right\}} \\
&= \sum_{n=0}^{+\infty} \mathbb{Q}(N = n) \cdot \frac{e^{\eta}}{\eta^n} \cdot \left(\int_0^T c(Y_s) ds \right)^n \\
&= \sum_{n=0}^{+\infty} e^{-\eta} \frac{\eta^n}{n!} \cdot \frac{e^{\eta}}{\eta^n} \cdot \left(\int_0^T c(Y_s) ds \right)^n \\
&= \sum_{n=0}^{+\infty} \frac{1}{n!} \left(\int_0^T c(Y_s) ds \right)^n \\
&= \exp \left(\int_0^T c(Y_s) ds \right)
\end{aligned}$$

With that in mind, let's look again at our

$$\mathbb{E}^{\mathbb{P}} \left[f(X_T) \mathbf{1}_{\tau_X > T} \right] = \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y_T)) \mathbf{1}_{\tau_Y > T} \cdot e^{h(Y_T) - h(y)} e^{\int_0^T c(Y_s) ds} \right]$$

in which we can substitute our new $e^{\int_0^T c(Y_s) ds}$ expression :

$$\begin{aligned}
& \mathbb{E}^{\mathbb{P}} \left[f(X_T) \mathbb{1}_{\tau_X > T} \right] \\
&= \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y_T)) \mathbb{1}_{\tau_Y > T} \cdot e^{h(Y_T) - h(y)} \mathbb{E}^{\mathbb{Q}} \left[e^{\eta \prod_{i=1}^N \left\{ \frac{T}{\eta} c(Y_{TU_i}) \right\}} \mid Y \right] \right] \\
&= \mathbb{E}^{\mathbb{Q}} \left[\mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y_T)) \mathbb{1}_{\tau_Y > T} \cdot e^{h(Y_T) - h(y)} e^{\eta \prod_{i=1}^N \left\{ \frac{T}{\eta} c(Y_{TU_i}) \right\}} \mid Y \right] \right]
\end{aligned}$$

Because $Y \mapsto f(g^{-1}(Y_T))$, $Y \mapsto \mathbb{1}_{\tau_Y > T}$ and $Y \mapsto h(Y_T)$ are measurable

$$= \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y_T)) \mathbb{1}_{\tau_Y > T} \cdot e^{h(Y_T) - h(y)} e^{\eta \prod_{i=1}^N \left\{ \frac{T}{\eta} c(Y_{TU_i}) \right\}} \right]$$

In conclusion, this formula concludes this first part :

$$\begin{aligned}
& \mathbb{E}^{\mathbb{P}} \left[f(X_T) \mathbb{1}_{\tau_X > T} \right] \\
&= \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y_T)) \mathbb{1}_{\tau_Y > T} \cdot e^{h(Y_T) - h(y)} e^{\eta \prod_{i=1}^N \left\{ \frac{T}{\eta} c(Y_{TU_i}) \right\}} \right]
\end{aligned}$$

2. Some additionnal work

Let's separate two cases : $\{\tau_X = 0\}$ and $\{\tau_X \geq 1\}$. For the convenience, we define :

$$Z := f(g^{-1}(Y_T))\mathbb{1}_{\tau_Y > T} \cdot e^{h(Y_T) - h(y)} e^\eta \prod_{i=1}^N \left\{ \frac{T}{\eta} c(Y_{TU_i}) \right\}$$

In the previous equation, it gives :

$$\begin{aligned} & \mathbb{E}^{\mathbb{P}}[f(X_T)\mathbb{1}_{\tau_X > T}] \\ &= \mathbb{E}^{\mathbb{Q}}[Z] = \mathbb{E}^{\mathbb{Q}}[\mathbb{E}^{\mathbb{Q}}[Z|N]] \\ &= \mathbb{E}^{\mathbb{Q}}[\mathbb{E}^{\mathbb{Q}}[Z|N=0] \cdot \mathbb{Q}(N=0) + \mathbb{E}^{\mathbb{Q}}[Z|N \geq 1] \cdot \mathbb{Q}(N \geq 1)] \\ &= \mathbb{E}^{\mathbb{Q}}[f(g^{-1}(Y_T))\mathbb{1}_{\tau_Y > T} \cdot e^{h(Y_T) - h(y)} e^\eta \cdot e^{-\eta}] \\ &\quad + \mathbb{E}^{\mathbb{Q}}[\mathbb{E}^{\mathbb{Q}}[Z|N \geq 1] \cdot \mathbb{Q}(N \geq 1)] \\ &= \mathbb{E}^{\mathbb{Q}}[f(g^{-1}(Y_T))\mathbb{1}_{\tau_Y > T} \cdot e^{h(Y_T) - h(y)}] \\ &\quad + \mathbb{E}^{\mathbb{Q}}[\mathbb{E}^{\mathbb{Q}}[Z \cdot \mathbb{1}_{N \geq 1}]] = C + \mathbb{E}^{\mathbb{Q}}[Z \cdot \mathbb{1}_{N \geq 1}] \end{aligned}$$

Let's dig deeper for C :

$$\begin{aligned} C &= \mathbb{E}^{\mathbb{Q}}[f(g^{-1}(Y_T))\mathbb{1}_{\tau_Y > T} e^{h(Y_T) - h(y)}] \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} f(g^{-1}(z))\mathbb{1}_{\tau > T} e^{h(z) - h(y)} p(\tau, y, z) dy d\tau \end{aligned}$$

where p is the joint density of (Y_T, τ_Y) conditionnally to $\{Y_0 = y\}$

$$\begin{aligned} &= \int_{\mathbb{R}} f(g^{-1}(z)) e^{h(z) - h(y)} \left(\int_{\mathbb{R}} \mathbb{1}_{\tau > T} \cdot p(\tau, y, z) d\tau \right) dy \\ &= \int_D f(g^{-1}(z)) e^{h(z) - h(y)} \left(\int_T^{+\infty} p(\tau, y, z) d\tau \right) dy \end{aligned}$$

because $p|_{z \notin D}(\tau, y, z) = 0$ (τ_Y is not finite for Y_T out of the barrier)

$$= \int_D f(g^{-1}(z)) e^{h(z) - h(y)} p^D(y, z) dy$$

where we define p^D as the restriction of p on $\{\tau > T\}$

At last we have

$$C = \int_D f \circ g^{-1}(z) e^{h(z)-h(y)} p^D(y, z) dz$$

$$Z = f(g^{-1}(Y_T)) \mathbb{1}_{\tau_Y > T} \cdot e^{h(Y_T)-h(y)} e^\eta \prod_{i=1}^N \left\{ \frac{T}{\eta} c(Y_{TU_i}) \right\}$$

$$\mathbb{E}^\mathbb{P} [f(X_T) \mathbb{1}_{\tau_X > T}] = C + \mathbb{E}^\mathbb{Q} [Z \cdot \mathbb{1}_{N \geq 1}]$$

3. Using $q(t, s, a, b)$ again

This step is crucial but quick thanks to our previous work in Chapter I, where we showed that, given the now ordered uniform points $U_{(i)}$, we had :

$$\mathbb{Q} \left(\min_{[0, T]} Y \geq g(L) | Y_T \right) = \prod_{i=1}^N \left\{ 1 - q \left(TU_{(i-1)}, TU_{(i)}, g^{-1}(Y_{TU_{(i-1)}}), g^{-1}(Y_{TU_{(i)}}) \right) \right\}$$

where :

- $q(t, s, a, b)$ was the probability for Y to go out of $[g(a), g(b)]$ between $[t, s]$
- $U_0 := 0$ and $U_N := 1$

But now, we know by definition of τ_Y that $\mathbb{Q} \left(\min_{[0, T]} Y \geq g(L) | Y_T \right) = \mathbb{Q}(\mathbb{1}_{\tau_Y > T} | Y_T)$ and we can define the new q^{BB} function by :

$$q^{\text{BB}}(\Delta, \alpha, \beta) := 1 - q(t, s, g(\alpha), g(\beta)) \text{ where } s - t = \Delta$$

And we get the following :

$$\tilde{Z} := f(g^{-1}(Y_T)) e^{h(Y_T)-h(y)} e^\eta \prod_{i=1}^N \left\{ \frac{T}{\eta} c(Y_{TU_{(i)}}) \right\}$$

$$\mathbb{E}^\mathbb{P} [f(X_T) \mathbb{1}_{\tau_X > T}] = C + \mathbb{E}^\mathbb{Q} \left[\tilde{Z} \cdot \mathbb{1}_{N \geq 1} \cdot q^{\text{BB}} \left(TU_{(i)} - TU_{(i-1)}, Y_{TU_{(i-1)}}, Y_{TU_{(i)}} \right) \right]$$